# Teaching Statement

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My goal as a teacher is to help students build the foundation they need to learn and appreciate mathematical ideas independently. This requires active engagement, so my teaching philosophy centers on *active learning*. Clear explanations matter, but actual learning occurs when students attempt problems themselves, try ideas and ask questions. I believe that an instructor's job is to provide this sort of guidance and opportunities to students because a method that looks clear on the board can feel quite different when a student has to decide what to do next.

At Florida State University, I have taught Precalculus Algebra, Calculus I, II, and III, and Applied Linear Algebra, and led recitations in Calculus I and Discrete Mathematics I. These experiences have shaped my approach around active learning, accessible explanations, and steady support inside and outside the classroom. My teaching has also been recognized through the Tam Family Award for Excellence in Teaching and Research and two nominations for the Outstanding Teaching Assistant Award in 2023–2024. Below, I describe several ways I put these principles into practice.

### Learning through examples and practice

Mathematics is usually presented in a definition-theorem-example sequence. I prefer to begin a new topic with accessible examples that motivate the formal ideas before a formal definition and theorem is presented. Connecting abstract statements to examples and intuition helps students build a mental picture of what they mean. As students become more familiar with the material, I gradually reduce my guidance and introduce more challenging problems. This scaffolding helps students build the confidence and skills to approach problems on their own. A Calculus II student described my instruction as "Very thorough instruction and used plenty of in-class examples in order to convey the course material."

In each class, I set aside about ten minutes for students to work on a problem related to the day's topic. For example, after discussing how to compute a determinant, I ask students to carry out the computation themselves. This gives them a chance to discover which steps they understand and where they need help. It also makes practice part of the lesson, while the ideas are still fresh. In my Summer 2026 Applied Linear Algebra course, I graded these activities for completion rather than correctness. One student described this as "a low-stress opportunity to practice the material." I want students to be willing to attempt a problem before they are certain of the answer.

In my experience, one of the best ways to understand mathematics is to try to explain it to someone else. I therefore give students opportunities to collaborate and learn from one another. In Applied Linear Algebra, students could work together on open-book, open-note quizzes. One student noted that this allowed classmates to help one another understand the material. These activities complement independent practice: explaining a method to a classmate encourages students to clarify their reasoning, identify gaps in their understanding, and consider why a solution works.

### Being approachable while building independence

Mathematics has a reputation for being difficult, and students may hesitate to share their reasoning or participate in class. During class, I make it clear that questions, comments, and interactions are welcome, and that mistakes are a normal part of learning. Through this positive attitude and encouragement, I hope to cultivate genuine intellectual curiosity and help students feel comfortable exploring ideas, offering their thoughts, and engaging with the material even when they are unsure. Interactions outside class are equally important: I regularly remind students about office hours and encourage them to email me or reach out when they need guidance. I want them to seek help early, rather than wait until they feel overwhelmed. One Calculus II student wrote, "He used a traditional method and made sure the class knew that they could ask any question they had after each example and explanation." Another noted,

"He genuinely treats his students with respect."

In my opinion, being supportive also means being honest about the effort a course requires. I often explain that passively listening to a solution is not enough: students need to make an active effort to explore ideas, ask questions, and practice. They need to invest time in solving problems and thinking about ideas to build their understanding and independence. The purpose of my support is to help them reach that independence. A student shared in an email, "Calculus has never been my strongest subject, but by the end of this semester it was my favorite subject and I appreciate all your support."

**Providing resources and room to improve**

I make lecture notes and other course materials available so that students can revisit an explanation or review something they missed. Before each exam, I provide study problems to help them identify what they understand and what needs more attention. These materials give students a clear path for preparation while leaving them responsible for doing the work. In my Calculus III evaluations, students highlighted the value of returning to notes from earlier lessons and using conceptual study questions to check their understanding. One student wrote, "Posting the notes after class and the study guides were always extremely helpful." My goal is for exam preparation to reinforce learning and help students connect ideas across the course.

I also believe that an early setback should leave room for improvement. Where appropriate, I offer additional opportunities for students to demonstrate what they have learned. For example, I have allowed the first midterm score to be replaced by the highest midterm score. This gives students who struggle at the beginning of a semester a reason to keep working, while still requiring them to demonstrate their understanding on a later exam. I have seen students recover from a difficult start and finish the semester strongly. I want assessment to recognize that growth and encourage continued effort. A Calculus II student wrote, "Although the course itself was difficult, Cagatay gave many opportunities to students to allow them to master the material and apply it to the tests."

**Learning from students and looking ahead**

Student feedback helps me understand both the strengths of my teaching and the areas that need attention. Students have valued my examples, practice activities, and availability, but some have also asked for a slower pace or more time on challenging topics. I take these comments seriously. In planning future courses, I will pay particular attention to the balance between explanation and practice, reserving time for difficult ideas and checking understanding before moving on.

Students in Applied Linear Algebra also asked for more examples connecting the theory to real applications. This is a direction I would like to develop further. My research in geometry, topological data analysis, and machine learning gives me opportunities to connect mathematical ideas to scientific questions. In future courses, I would like to introduce small data-based or computational examples that fit the students' preparation and the course goals.

One Calculus II student wrote, "I feel that I actually learned material instead of just learning how to pass the class this semester." That is exactly the outcome I hope to support: students who appreciate learning the material rather than merely passing the course.